Master Technical Plan • Duration: 15 Days Max • 100% AI Authored & Automated

Daksh — Complete 15-Day Engineering & Testing Plan

The entire Daksh BaaS suite—Multiplayer Netcode, Analytics Telemetry, Economy V2, AI Command Menu, Dual Payments, and Helm Architecture—delivered and verified with zero human coding or manual QA.

Multiplayer Trifecta | ClickHouse Telemetry | PlayFab Economy V2 | Razorpay + Paddle | Helm Only | AI Automated Tests

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> **Operating Model:** We do not write code or test cases manually. Claude writes the backend netcode and distributed algorithms, Codex writes the Unity C# SDK and unit tests, and Antigravity orchestrates the local filesystem, compiles Next.js builds, and runs the automated test runners. Every single day ships both the production code and its automated test pass.

Phase 1: Local Foundation, Identity & Economy

DAYS 1 TO 5

Day 01 Containerized Foundation & `npx daksh dev` CLI

Claude + Codex + Antigravity

AI sets up the Docker Compose stack (Nakama 3.22, PostgreSQL 16, Redis 7) and writes the 1-command CLI that boots local containers and launches the studio in under 60 seconds.

> ✓ Automated test: Vitest runs the CLI in a child process; verifies port 7350 and 3000 respond with HTTP 200 in <15 seconds.

Day 02 Player Identity Federation & Device Auth

Claude + Codex

AI writes zero-friction device ID authentication and multi-provider identity linking (Steam, Google Play, Apple) with Ed25519 JWT session tokens.

> ✓ Automated test: Generates 100 random player sessions; verifies signature validation, token expiry, and token refresh.

Day 03 PlayFab Economy V2 (Currencies & Catalogs)

Claude

AI writes the Nakama TypeScript V8 runtime extension for virtual currencies (Gold/Gems), item catalogs, item bundles, and player inventory balances.

> ✓ Automated test: Runs concurrent buy operations with row-level database locks (FOR UPDATE) to assert zero wallet balance race conditions.

Day 04 Loot Drop Tables & Idempotency Engine

Claude + Codex

AI writes weighted random loot tables for mystery boxes and sets up the transaction deduplication engine to protect players on flaky mobile networks.

> ✓ Automated test: Fires 50 parallel duplicate purchase requests; asserts 1 succeeds, 49 are safely ignored, and RNG seed distribution is uniform.

Day 05 AI Command Menu (Cmd+K) & MCP Server

Antigravity + Codex

AI builds the global command palette in the Next.js studio and open MCP server, letting designers create complete catalogs via natural language in seconds.

> ✓ Automated test: Feeds 20 diverse prompts; asserts schema generator produces valid JSON and dispatches studio UI reloads in <1.5 seconds.

Phase 2: Complete Multiplayer Netcode & Analytics

DAYS 6 TO 10

Day 06 Turn-Based Multiplayer Engine (Async & Sync)

Claude

AI writes the turn-based room state handlers for board, puzzle, and card games (8-Ball Pool, Chess, Poker) running in Nakama memory with $0 compute overhead.

> ✓ Automated test: Simulates a 2-player game loop: turn handoffs, turn timeouts, surrender events, and atomic commit of final match winnings to PostgreSQL.

Day 07 Real-Time Relay Netcode (UDP & WebSockets)

Claude

AI writes the client-authoritative packet relay engine for high-speed action and racing games, supporting up to 64 concurrent players per match room.

> ✓ Automated test: Connects 32 mock WebSocket clients; broadcasts continuous position packets and verifies <10ms server relay latency jitter.

Day 08 AccelByte-Grade Matchmaking & Dynamic Rule Flexing

Claude + Codex

AI implements customizable matchmaking rulesets: latency ping limits, rank skill delta (MMR), and automated rule relaxation if a player waits >15 seconds.

> ✓ Automated test: Simulates 100 queued players across 3 regions; confirms strict matching early, followed by verified rule flexing for long-waiting players.

Day 09 Matchmaking X-Ray Admin Portal & Leaderboards

Antigravity + Codex

AI builds the visual Matchmaking X-Ray dashboard tab to inspect live match queues and builds real-time Redis leaderboards with tiered seasonal resets.

✓ Automated test: Injects 500 player scores; verifies sorted set order, tie-breaking logic, and live queue telemetry feed in the studio UI.

Day 10 High-Volume Event Telemetry & Analytics Pipeline

Claude + Antigravity

AI builds the game event ingestion endpoint (`/api/analytics/events`) buffering player kills, loot pickups, purchases, and session durations into ClickHouse/Postgres.

✓ Automated test: Flushes a batch of 10,000 mock gameplay events; confirms zero data loss, batch buffer flush in <100ms, and partitioned table indexing.

Phase 3: Real-Time KPIs, Payments, SDK & Launch

DAYS 11 TO 15

Day 11 Real-Time Analytics Dashboards (DAU, Retention, ARPU)

Codex + Antigravity

AI builds the visual studio analytics view calculating Daily Active Users (DAU), Monthly Active Users (MAU), D1/D7/D30 Retention curves, and Average Revenue Per User.

✓ Automated test: Runs analytical SQL aggregations over 100,000 historical events; verifies accuracy of calculated retention and churn curves.

Day 12 Dual Payment Engine (Razorpay India + Paddle Global MoR)

Claude + Antigravity

AI sets up the geo-routed checkout endpoint: Indian developers pay via Razorpay (UPI, RuPay in INR), while international developers pay via Paddle (USD cards with automated VAT remittance).

✓ Automated test: Verifies HMAC-SHA256 signature verification for Razorpay and Paddle webhooks; confirms title entitlement flips to 'pro'.

Day 13 Unity C# SDK (UPM) & Offline Persistent Disk Queue

Codex

AI packages the official Unity client (`com.daksh.sdk`) with `DakshClient`, Auth, Economy, Netcode, and a local persistent SQLite/JSON retry queue for dropped connections.

✓ Automated test: Runs 24 C# NUnit tests; simulates dropped Wi-Fi, queues 10 actions to disk, reconnects, and asserts zero lost transactions.

Day 14 Helm Chart Only & The Famous Eject Button

Claude + Antigravity

AI writes the production Helm chart in `deploy/helm/daksh` with 100% parameterized `values.yaml` and builds the 1-click self-hosting zip export containing the Helm chart and database dump.

✓ Automated test: Runs Helm dry-run template linting and unpacks the exported zip on an isolated sandbox VM to confirm independent boot.

Day 15 Air-Gapped Flight Mode Test & Public Release

`All AI Agents`

AI disables Wi-Fi completely on the host machine, executes the complete game loop offline, runs a 5,000 CCU localhost stress test, compiles all 22 Next.js routes, and creates Git tag `v1.0.0`.

> ✓ Automated test: Master CI runner verifies 100% test pass with zero internet, sub-20ms latency under 5,000 CCU, and 0 TypeScript build errors.

Summary of the Full-Suite Deliverables

CATEGORY	WHAT AI IMPLEMENTS	AUTOMATED TEST STRATEGY
Multiplayer Netcode	Turn-Based (Pool/Chess), Real-Time Relay (WS/UDP), Matchmaking Rulesets, Matchmaking X-Ray.	32-client socket relay latency tests, turn-timeout win assertions, MMR flex tests.
Analytics & Telemetry	High-volume event ingestion, ClickHouse/Postgres OLAP, DAU/MAU, Retention D1-D30, ARPU curves.	10,000 events/sec batch ingestion test, analytical retention calculation checks.
Economy & LiveOps	Currencies (Gold/Gems), Catalogs, Stores, Drop Tables, AI Command Menu (`Cmd+K`).	Row-level lock concurrency tests (`FOR UPDATE`), 50 duplicate request deduplication.
Dual Payments	Razorpay (India UPI/RuPay in INR) + Paddle (Global MoR in USD with EU/UK VAT handling).	HMAC-SHA256 signature verification tests, country header geo-routing mocks.
Client & Deploy	Unity C# UPM SDK with offline retry queue; Helm Chart (`deploy/helm/daksh`); 1-Click Eject.	24 C# NUnit tests, Helm template dry-runs, 0-WiFi flight mode verification.
